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MANAGEMENT REVIEW

Management without theatre.



KPIS

QUALITY: FIRST PASS YIELD AND THE COST OF QUALITY

Where the idea of a cost of quality came from, how much poor quality costs, the four groups of cost, whether quality is still free, and a card for First Pass Yield.

15–20%

of sales revenue is what quality-related costs reach in many organisations, by ASQ's estimate; in some, as much as 40% of operations.

ASQ

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NO. 22 · KPIS

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Every operation pays for its mistakes: the order packed twice, the delivery made again, the complaint answered. How much of that cost can an operation actually see, and which measure shows it earliest?

Joseph Juran's handbook of 1951 first treated quality costs and called the cost of failures gold in the mine. Armand Feigenbaum split them into four groups in 1956; in 1979 Philip Crosby argued that quality is free and that what costs money is not doing a job right the first time. ASQ estimates that in many organisations these costs reach 15–20% of sales. First Pass Yield shows, step by step, how much work goes right the first time.

Stiven Janaqi EDITOR · STIVEN CATALYST

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HOW TO READ THIS ISSUE

Figure
Every figure has its source and year at the foot of its page.

Our reading
Where the editor interprets rather than the research, it says so.

Practice
The steps and the card are proposals to try, not research results.

COVER STORY

Gold in the mine

The idea that quality has a cost that can be counted is older than most quality programmes. It took shape between 1951 and 1979, in a handbook, an article and a book.

1951

Juran's Quality-Control Handbook first treats quality costs and calls the cost of failures gold in the mine.

1956

Feigenbaum, in Harvard Business Review, splits them into four groups: prevention, appraisal, internal and external failure.

1979

Crosby, Quality Is Free: quality is conformance to requirements, measured by the price of nonconformance.

“What costs money are the unquality things — all the actions that involve not doing jobs right the first time.”

Juran's point was that failure costs can be cut sharply by investing in improvement, which makes them a source of money as well as a loss. Crosby went further: the money is in the price of not conforming.

OUR READING

The cost of poor quality appears on no single line of the accounts. It is spread across rework, extra checks, returns and the time of the people who put things right.

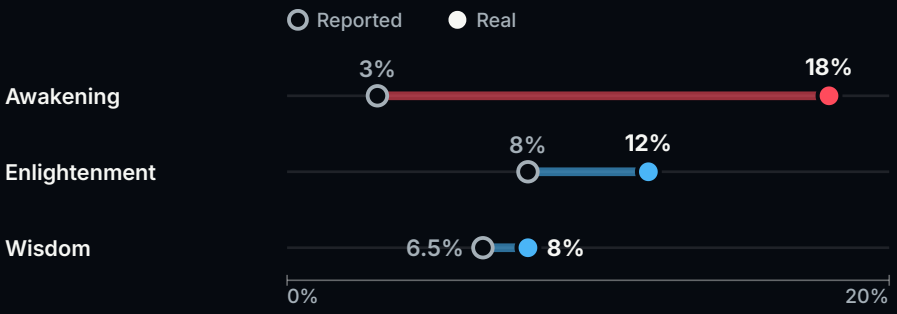
Sources: Joseph M. Juran (ed.), McGraw-Hill, 1951 (via Secondary summaries (Garvin)); Armand V. Feigenbaum, Harvard Business Review, 1956 (via Secondary summaries); Philip B. Crosby, McGraw-Hill, 1979 (via Reproductions of the maturity grid and summaries)

THE NUMBERS

What poor quality costs

ASQ estimates that in many organisations the true quality-related costs reach 15–20% of sales revenue, and in some as much as 40% of total operations. It gives no method, so the figure is an estimate rather than a measurement.

COST OF QUALITY AS A SHARE OF SALES IN CROSBY'S MATURITY GRID, BY STAGE



In his maturity grid of 1979, Crosby set the cost of quality that companies report against what he estimated it really is. In the first stage, uncertainty, the cost is unknown; he put it at 20% of sales. In the last, certainty, at 2.5%.

The cost shows outside the factory too: an OECD review of the evidence found that 15% of hospital spending and activity goes on treating failures of patient safety.

OUR READING

The reported cost of quality is whatever someone has counted. In your own operation, who counts the hours that go into rework?

Sources: ASQ, 2026 (via Quoted by ETQ and others); Philip B. Crosby, McGraw-Hill, 1979 (via Reproductions of the maturity grid and summaries); Luke Slawomirski, Ane Auraaen & Niek Klazinga, OECD, 2017

Crosby's figures are illustrative estimates from 1979; nobody measured them.

THE MODEL

Four groups of cost

The model Feigenbaum described in 1956 is still the most widely used: two groups of cost are spent to get quality right, two are paid when it goes wrong.

The cost of good quality

Prevention: training, process design, keeping the quality system

Appraisal: measuring, inspecting, testing, auditing

The cost of poor quality

Internal failure, found before the customer: scrap, rework, failure analysis

External failure, found by the customer: returns, warranty, complaints

HYPOTHETICAL EXAMPLE, A WAREHOUSE

Prevention	Training on the new labels
Appraisal	Scanning every pallet before loading
Internal	Repacking a wrong order before it leaves
External	A second delivery after a complaint

The later a mistake is found, the more steps and people it has already passed through. The examples are made up.

OUR READING

A team that counts only failures sees half the model. The other half is where the money for fewer failures gets spent.

Sources: Armand V. Feigenbaum, Harvard Business Review, 1956 (via Secondary summaries); ASQ, 2026 (via Quoted by ETQ and others)

The four groups are Feigenbaum's, as ASQ describes them today; the examples are the editors'.

More in the essay KPIs the team trusts

WHAT THE RESEARCH SAYS

Is quality still free?

Two studies in the International Journal of Quality & Reliability Management look at how the cost of quality is used, and at how it varies with the level of quality.

01	**Most programmes do not use it**	Reviewing the research on cost-of-quality models, Schiffauerova and Thomson found that most quality programmes do not use the cost of quality. Prevention–appraisal–failure is the most used model, and companies that use one report lower quality costs and better quality for their customers.
02	**Higher quality, lower total cost**	With data from manufacturing, Plewa, Kaiser and Hartmann found that at higher quality the total cost of quality and the cost of failures are significantly lower, while prevention and appraisal spending is not significantly higher.

OUR READING

The research supports Crosby more than it contradicts him: better quality went with lower total cost. It shows an association, though, not a recipe.

Sources: A. Schiffauerova & V. Thomson, International Journal of Quality & Reliability Management, 2006 (via Abstract); M. Plewa, G. Kaiser & E. Hartmann, International Journal of Quality & Reliability Management, 2016 (via Abstract)

Both rest on reported cases or correlations; we could not see the sample size of the 2016 study.

HOW IT IS MEASURED

First Pass Yield, step by step

First Pass Yield is the share of units that pass a step the first time, without rework or scrap. Multiplied across all the steps, it gives the Rolled Throughput Yield: the chance that a unit passes the whole process without a defect.

1. **List the steps**
   From order to delivery, in the order the work happens.
2. **Count the first-time passes**
   A unit that was reworked does not count as good.
3. **Multiply**
   Four steps at 85% each give 52.2%: about half the units pass every step the first time.
4. **Start with the lowest step**
   Improve the step with the lowest yield first.

HYPOTHETICAL EXAMPLE, AN ORDER THROUGH A WAREHOUSE

Picking	97 of 100 right the first time
Packing	99 of 100
Loading	98 of 100
Delivery	96 of 100

Each step looks good, yet only 90.3% of orders pass all four the first time. These numbers are invented.

Source: iSixSigma, 2026

The editors wrote the steps and the example; the formula follows iSixSigma.

TOOL OF THE ISSUE

The First Pass Yield card

One card for one process. Fill in a week of counts, multiply, and choose the step to improve first.

01 Process from where to where, and which unit is counted

02 Steps and First Pass Yield step · units in · right the first time · share

03 Rolled Throughput Yield the shares of all the steps, multiplied

04 Weakest step the lowest share, and its most frequent failure

05 Where it is paid internal failure or external failure

06 One change what we change this week, and when we count again

Sources: Armand V. Feigenbaum, Harvard Business Review, 1956 (via Secondary summaries); iSixSigma, 2026

The editors' card, building on Feigenbaum's groups of cost and the yields as iSixSigma defines them.

Open the tool: Damage Control stivencatalyst.com/tools/damage-control/

SOURCES AND METHOD

Every figure has a source.

The figures in this issue come from the sources below. The year shows how recent each one is.

Quality-Control Handbook
Joseph M. Juran (ed.), McGraw-Hill, 1951 · via Secondary summaries (Garvin)

Total Quality Control
Armand V. Feigenbaum, Harvard Business Review, 1956 · via Secondary summaries

Quality Is Free: The Art of Making Quality Certain
Philip B. Crosby, McGraw-Hill, 1979 · via Reproductions of the maturity grid and summaries

What is Cost of Quality (COQ)?
ASQ, 2026 · via Quoted by ETQ and others
https://asq.org/quality-resources/cost-of-quality

The economics of patient safety: Strengthening a value-based approach to reducing patient harm at national level
Luke Slawomirski, Ane Auraaen & Niek Klazinga, OECD, 2017
https://doi.org/10.1787/5a9858cd-en

A review of research on cost of quality models and best practices
A. Schiffauerova & V. Thomson, International Journal of Quality & Reliability Management, 2006 · via Abstract
https://doi.org/10.1108/02656710610672470

Is quality still free? Empirical evidence on quality cost in modern manufacturing
M. Plewa, G. Kaiser & E. Hartmann, International Journal of Quality & Reliability Management, 2016 · via Abstract
https://doi.org/10.1108/IJQRM-11-2014-0189

What is rolled throughput yield?
iSixSigma, 2026
https://www.isixsigma.com/dictionary/rolled-throughput-yield-rty/

EDITORIAL METHOD

Each figure was checked for its year, its publisher and what it measures, in the full text where it could be opened. A source read only in a summary, an abstract or excerpts is marked “via”. A figure not confirmed in what was read is marked so on its page, or left out. The editor's interpretation is marked “Our reading”.



MANAGEMENT REVIEW

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Every issue, one management question, checked against the best research.

ALL ISSUES

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Edited by Stiven Janaqi, who answers for it; “the editors” in this publication means him. Its texts, translations, layout and the checking of figures against their sources are done in working sessions with Claude, an AI model by Anthropic.
stivencatalyst.com/how-this-is-made.html